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EE2006

I) magnetic circuits:

i) $\underset{mmf}{Ni} = \oint \bar{H}.d\bar{l} = Hm Lm$

ii) $\phi = \bar{B}.\bar{A} = \mu H$ (flux)

iii) $B = \mu H$ (B → flux density)

iv) $\phi = \frac{Ni}{(l/\mu A) \rightarrow \text{Reluctance } (\mathcal{R})}$ ϕ → equivalent to current

ϕ vs I linear & saturates (knee point)

v) $\lambda = LI$ (flux linkage) $= N\phi$

$L = \frac{\lambda}{I} = \frac{N\phi}{I} = \frac{N(B.A)}{Hl/N} = \frac{N^2}{\mathcal{R}}$

vi) air gap ⇒ $Ni = Hc lc + Hg lg$ (no flux leakage)

$Ni = \frac{Bc lc}{\mu_0 \mu_r} + \frac{Bg lg}{\mu_0} = \phi(\mathcal{R}c + \mathcal{R}g)$

vii) $e = \frac{d\lambda}{dt} = Bm a w \sin(wt + \pi/2) = Blv$

statically induced ↓ dynamically induced

viii) loss ∝ area under B-H curve

II) Transformers

i) $\phi \propto \frac{E}{f}$ $\frac{E_1}{E_2} = \frac{N_1}{N_2} = \frac{I_2}{I_1} = a$

ii) $e = 2\pi f \phi_m N_1$ $e_{rms} = \sqrt{2}\pi f \phi_m N_1$

iii) $I_2 = 0 \Rightarrow$ Eddy losses i.e eddy + hysterisis

iv) 1° side $R_1 =$ 2° side $\frac{R_1}{a^2} = R_1'$

2° side $R_2 =$ 1° side $R_1 a^2 = R_2'$

v) $\eta = \frac{P_{out}}{P_{in}}$ for a given load

vi) $V.R = \frac{V_{20} - V_{2L}}{V_{2L}} \times 100$ $V_{20} \rightarrow 2^\circ$ terminal voltage @ 0 load

$V_{2L} \rightarrow 2^\circ$ terminal voltage @ full load

vii) I/P power − O/P power → core loss + Cu-loss

↓ const for given f ↓ varies with load

η is max when $P_{cu} = P_{fe}$

viii) OC test ⇒ R_c, x_m SC test ⇒ R_{eq}, x_{eq}

ix) Power $= VI^*$ = Real power + reactive power

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III DC machines carries Id current / induced emf

i) (a) rotor = Armature ↗ wave → high V low I
(b) Stator = field → establishes & maintains mag fields lap → low V high I

ii) $E_{coil} = \frac{\phi P \omega_m}{2\pi}$ P → poles
ϕ → flux per pole

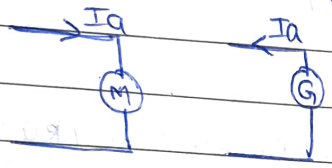
$E = K_e \phi \omega_m$ ϕ → controlled by field current I_f

iii) commutator → rectifier / inverter

iv) $T = K_t \phi I_a$ No load ⇒ $I_a = 0$

v) $V_{in} = E_b + I_a R_a$ $E_b = K_e \phi \omega = \frac{P \phi N Z}{60 A}$

vi)



$V_{in} = E_b + I_a R_a$ + brushdrop $E_g = V_L + I_a R_a$ + brushdrop

vii) rated power: Pmax to load

viii) $\omega_m = \frac{V}{K_e \phi} - \frac{I_a R_a}{(K_e \phi)} = \underbrace{\frac{V}{K_e \phi}}_{\omega_{mo}} - \frac{R_a T_e}{(K_e \phi)^2}$

ix) $\underbrace{VI_a}_{P_{in}} = \underbrace{E_b I_a}_{P_{out}} + \underbrace{R_a I_a^2}_{\text{arm loss}}$

IV Synchronous machines:

i) $\vec{E_f} = \vec{V_t} + \vec{I_a}\vec{Z_s}$ ~~MOTOR~~ generator; $\vec{V} = \vec{E} + \vec{I_a}\vec{Z_s}$ motor

ii) $P = \frac{EV}{X_S} \sin\delta$ 3phase power = 3P

$T = \frac{P}{\omega_s}$ $\cos\phi = \frac{P}{\sqrt{3} VI}$ $\delta = \angle E - \angle V$ → full load torque angle

$P_{cu} = 3I^2 R_a$ $Q = 3V/X_S (E\cos\delta - V)$ $\frac{E}{V} \propto f_s$

iii) $N_s = \frac{120 f}{P}$

iv) $E_\phi = \sqrt{(V\phi \cos\phi + I_a R_a)^2 + (V\phi \sin\phi + I_a X_s)^2}$

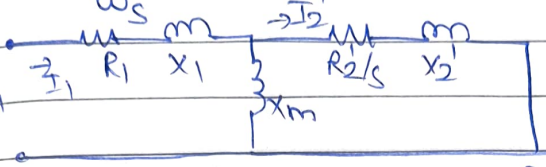
v) $E_{ph} = 4.44 f \phi T_{ph} \cdot K_w$

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V) Induction machines

(i) $s = \frac{\omega_s - \omega_m}{\omega_s}$ $\quad \because f_r = sf$

(ii)



$\frac{R_2'}{s} = R_2' + R_2'\left(\frac{1-s}{s}\right)$
$\downarrow$
P_{cu}

$P_g = 3I_2'^2 \frac{R_2'}{s}$ $\quad P_{rcu} = 3I_2'^2 R_2'$ $\quad P_m = P_g - P_{r,cu}$

$T_e = \frac{P_m}{\omega_m}$ $\quad T = \frac{P_{ag}}{\omega_s}$ (induced)

$T \propto s$ $\quad I_2' = \frac{V_1}{(R_1 + R_2'/s) + j(X_1 + X_2')}$ $\quad$ Shaft $T = \frac{P_o}{\omega_r}$

(iii) max torque at $s = \frac{R_2'}{(R_1^2 + (X_1 + X_2')^2)^{1/2}}$

(iv) ~~Blocked rotor~~ S.C test $\Rightarrow s = 1$ (Blocked rotor)

(v) $P_{in} = \sqrt{3} V_L I_L \cos\phi$

$P_{ag} = P_{in} - P_{scl} - P_{core}$ $\quad P_{ag} : P_{rcl} : P_{mech} = 1 : s : 1-s$

$\hookrightarrow 3I_1^2 R_1$

$\hookrightarrow 3I_2'^2 R_2'/s$

$P_o = P_{conv} - P_{mech}$